

(Hyper)elliptic Riemann Surfaces

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Abstract

This is a quick introduction to a special class of compact Riemann surfaces: the so called (hyper)elliptic curves. Most of the text is based on the lectures from a course on Riemann Surfaces, held at IMPA in 2025 by Jorge Vitório Pereira. His lecture notes can be found in his website. We begin with an elementary approach to elliptic curves and then proceed to more abstract results concerning the hyperelliptic ones. The main goal is to prove a criterion for hyperellipticity. Of course, much more can be said about hyperelliptic surfaces, but that is beyond the scope of this text. From section 2 until the end, the reader is expected to have some familiarity with sheaves and cohomology. For the interested reader, [1, 2, 5] are good references.

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1 Elliptic curves

1.1 Introduction

Consider the elementary integral

$$\int_0^1 (1-x^2)^{-1/2} dx = \int_0^{\pi/2} (1-\sin^2\theta)^{-1/2} \cos\theta d\theta = \pi/2 .$$

Its seemingly innocent variation

$$\int_0^1 (1-x^4)^{-1/2} dx$$

turns out to be mysteriously too complicated for a regular calculus exercise. Spoiler: we are dealing with a (complete) *elliptic integral*, which is the integral of a rational function of x and y such that y^2 is a polynomial in x , of degree 3 or 4, with simple roots. These were studied extensively by Legendre, Gauss, Abel and Jacobi throughout the 19th century [3]. It turns out that elliptic integrals are inverted by *elliptic functions*, which have a fundamental role on the theory of *elliptic curves* [4].

1.2 Elliptic Functions

Definition 1.1. *The set $\mathcal{C} = \{[X : Y : Z] \in \mathbb{CP}^2 \mid Y^2Z = X^3 + aXZ + bZ\}$ with $a, b \in \mathbb{C}$ is called an elliptic curve (over the field $\mathbb{C}$).*

Remark 1.1. *An elliptic curve is the union of its affine part (i.e. points of $\mathcal{C} \subset \mathbb{CP}^2$ such that $Z \neq 0$) with a unique point "at infinity", $\mathcal{O} = [0 : 1 : 0]$, which is the intersection of $\mathcal{C}$ with the "line at infinity", $\{Z = 0\}$.*

Remark 1.2. *Given two points on an elliptic curve $p, q \in \mathcal{C}$, one can obtain another one, by intersecting the line passing through p and q with $\mathcal{C}$. This is known as the chord-and-tangent law and it endows an abelian group structure on (the rational points of) $\mathcal{C}$. We denote the group associated to $\mathcal{C}$ by E from now on. This is because if the underlying field were not algebraically closed, one would have $E \subset \mathcal{C}$. Observe that the point at infinity $\mathcal{O}$ is the identity of the group.*

The following theorem is a celebrated result on the arithmetic of elliptic curves. A proof can be found in [3].

Theorem 1.1 (Mordell-Weil). *The group E of K -rational points of an elliptic curve over a field K is finitely generated.*

We now establish the relation between elliptic functions and elliptic curves. The key concept here is periodicity. Observe that a function of a real variable cannot have two independent periods, say ω_1 and ω_2 . Indeed, if $\omega_1/\omega_2 \in \mathbb{Q}$, then there is essentially one period: their greatest common divisor. If the quotient is not rational, then the periods are dense, which imply that the function must be constant, if we require continuity. Thus, we are interested in functions of one complex variable $f(z)$, $z \in \mathbb{C}$ such that $f(z + \omega_1) = f(z)$ and $f(z + \omega_2) = f(z)$, with $\omega_1/\omega_2 \notin \mathbb{R}$. We say that such f is Λ -periodic, where $\Lambda = \{n\omega_1 + m\omega_2 \in \mathbb{C} \mid n, m \in \mathbb{Z}\}$ is the lattice generated by the periods ω_1 and ω_2 .

Theorem 1.2. *Let f be Λ -periodic. Then:*

- *If f is holomorphic, then it is constant;*
- *Let $\mathcal{P}$ be the fundamental domain of Λ . The following properties are satisfied:*
 - *With multiplicities counted, f has the same number of zeros and poles on $\mathcal{P}$.*
 - *The sum of the residues of f on $\mathcal{P}$ is zero.*
 - *The difference between the sum of zeros and the sum of poles (with multiplicities counted) of f is zero mod Λ .*

Proof. Since $\mathcal{P}$ is compact and f continuous, it must be bounded on $\mathcal{P}$. But periodicity implies that f is bounded on the whole plane, so it must be constant by Liouville's theorem.

Notice we can assume that no zeros nor poles of $f(z)$ and $f'(z)$ lie on the boundary $\partial\mathcal{P}$, by translating by a constant. Thus, the difference of zeros and poles of f on $\mathcal{P}$ is

$$\frac{1}{2\pi i} \int_{\partial\mathcal{P}} \frac{f'(z)}{f(z)} dz ,$$

which is zero, since the opposite sides of $\partial\mathcal{P}$ cancel by periodicity.

To see that the sum of the residues is zero, apply the same argument as above, but integrating $f(z)$ instead.

Finally, the difference $S := (f)_0 - (f)_\infty$ of zeros and poles of f is given by

$$S = \frac{1}{2\pi i} \int_{\partial\mathcal{P}} \frac{zf'(z)}{f(z)} dz .$$

Let L_1 and L_2 be the line segments from zero to ω_1 and ω_2 in the complex plane, respectively. Notice that $\partial\mathcal{P} = L_1 \cup (L_1 + \omega_2) \cup L_2 \cup (L_2 + \omega_1)$. Choosing anti-clockwise orientation and again by periodicity, we have

$$\begin{aligned} S &= \frac{1}{2\pi i} \left(\int_{L_1} \frac{zf'(z)}{f(z)} dz - \int_{L_1} \frac{(z + \omega_2)f'(z + \omega_2)}{f(z + \omega_2)} dz - \right. \\ &\quad \left. \int_{L_2} \frac{zf'(z)}{f(z)} dz + \int_{L_2} \frac{(z + \omega_1)f'(z + \omega_1)}{f(z + \omega_1)} dz \right) \\ &= \frac{1}{2\pi i} \left(\omega_1 \int_{L_2} \frac{f'(z)}{f(z)} dz - \omega_2 \int_{L_1} \frac{f'(z)}{f(z)} dz \right) . \end{aligned}$$

Using the change of variables $\xi = f(z)$ and observing that $f(L_i)$ is a loop by periodicity, one gets

$$S = \frac{1}{2\pi i} \left(\omega_1 \int_{f(L_2)} \frac{d\xi}{\xi} - \omega_2 \int_{f(L_1)} \frac{d\xi}{\xi} \right) \in \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 ,$$

as wanted. □

Remark 1.3. *From the previous theorem, a Λ -periodic function must be meromorphic with poles of order higher than 1. We are going to define such a function explicitly.*

Definition 1.2. *The Weierstrass $\wp$ -function associated to the lattice Λ is defined by*

$$\wp(z) := \frac{1}{z^2} + \sum_{\omega \in \Lambda - \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right) .$$

The series converges absolutely on a disk of arbitrary radius for all $z \notin \Lambda$. We show that $\wp$ is indeed Λ -periodic. Notice that

$$\wp'(z) = - \sum_{\omega \in \Lambda} \frac{2}{(z - \omega)^3} ,$$

which is Λ -periodic by the following general argument: Suppose we have a group G acting on a vector space V . For any $v \in V$, $\tilde{v} := \sum_{g \in G} g \cdot v$ is fixed by G if and only if this

series converges. In our case, we have $G = \Lambda$ acting by translation on the vector space of meromorphic functions on $\mathbb{C}$, and for $v = f(z) = 1/z^3$, the associated series converges, so it is fixed by the group action, which means it is Λ -periodic.

Now let $f(z) := \wp(z + \omega_1) - \wp(z)$. Clearly, $f'(z) = 0$, so f is constant. But since $\wp$ is (clearly) an even function, we have

$$f(-\frac{\omega_1}{2}) = \wp(\frac{\omega_1}{2}) - \wp(-\frac{\omega_1}{2}) = 0 ,$$

and so $\wp(z + \omega_1) = \wp(z)$. The same argument applied to the other period proves the claim.

1.3 The Isomorphism

Theorem 1.3. *The function $\wp$ satisfies the differential equation*

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3 ,$$

where $g_2 = 60 \sum_{\omega \in \Lambda - \{0\}} \frac{1}{\omega^4}$ and $g_3 = 140 \sum_{\omega \in \Lambda - \{0\}} \frac{1}{\omega^6}$ are constants that depend on the lattice Λ .

Corollary 1.1. *For any $z \in \mathbb{C}$, the (affine) point $(\wp(z), \wp'(z)) \in \mathbb{C}^2$ belongs to the elliptic curve given by equation $y^2 = 4x^3 - g_2x - g_3$.*

The moral is that we may construct a map from any complex (one dimensional) torus to its associated elliptic curve via the elliptic function $\wp$:

$$\mathbb{C}/\Lambda \rightarrow \mathcal{C} , \quad z(\text{mod } \Lambda) \mapsto [z^3\wp(z) : z^3\wp'(z) : z^3]$$

In fact, the map above is an isomorphism of abelian groups!

Theorem 1.4 (Uniformization). *Let $a, b \in \mathbb{C}$ be such that $a^3 - 27b^2 \neq 0$. There is a unique lattice Λ for which $g_2(\Lambda) = a$ and $g_3(\Lambda) = b$. In other words, any non-singular elliptic curve can be seen as a genus 1 Riemann surface for an appropriate lattice.*

An elliptic curve (over $\mathbb{C}$) can be defined alternatively as a compact Riemann surface X of genus 1, together with a choice of a point $p \in X$, that plays the role of the identity element of the group [5].

2 Linear Systems and Projective Embeddings

In order to grasp the concept of (hyper)elliptic Riemann surfaces, one must understand how they can be embedded in projective spaces. This is the goal of this section. We present a correspondence between maps from a (compact) Riemann surface X to $\mathbb{CP}^r$ of degree d and the so called *linear systems* g_d^r , defined below. Some details are omitted, but can be found for example in [2].

We begin by setting the notation: For a holomorphic line bundle L over a Riemann surface X , we denote by $\mathcal{L}$ its sheaf of sections and by $H^k(X, \mathcal{L})$ the k -th Čech cohomology group of X with coefficients on $\mathcal{L}$. Also, set $h^k = \dim H^k$. The canonical divisor and the sheaf of holomorphic differentials on X are denoted K_X and Ω_X^1 , respectively. Recall that $\mathcal{L} = \mathcal{O}_X(D)$: The sub-sheaf of meromorphic functions on X whose associated divisor is $\geq -D$, where $D \in \text{Div}(X)$ is the divisor associated to any non-zero meromorphic section of $\mathcal{L}$. For convenience, we also recall the Riemann-Roch formula, which will be used extensively. A proof can be found in [1].

Theorem 2.1 (Riemann-Roch). *Let X be a compact Riemann surface of genus g . For any divisor $D \in \text{Div}(X)$,*

$$h^0(X, \mathcal{O}_X(D)) - h^0(X, \mathcal{O}_X(K_X - D)) = \deg(D) + 1 - g .$$

Definition 2.1. *A linear system of degree d and rank r on a compact Riemann surface X is a pair (L, V) , where L is a holomorphic line bundle of degree d over X and $V \subset H^0(X, \mathcal{L})$ is a subvectorspace of dimension $r + 1$. It is often denoted by g_d^r .*

Definition 2.2. *A linear system (L, V) is complete if it has maximal dimension. i.e., $V = H^0(X, \mathcal{L})$. The complete linear system associated to the holomorphic line bundle L is denoted by $|L|$ (or $|D|$).*

Definition 2.3. *A base point $p \in X$ of a linear system (L, V) is a point on which all the elements of V vanish.*

We now sketch the correspondence between linear systems and applications to $\mathbb{CP}^r$:

Let $(L, V) = g_d^r$ be a base point free linear system on X . Choose a base $\{\sigma_0, \dots, \sigma_r\}$ of V and define the map

$$\Phi : X \rightarrow \mathbb{CP}^r , \quad p \mapsto [\sigma_0(p) : \dots : \sigma_r(p)] .$$

The meaning of $[\sigma_0(p) : \dots : \sigma_r(p)]$ is the following: for each point $p \in X$, let z be a local coordinate so that $\sigma_i = f_i(z)\sigma$, where f_i is holomorphic on a neighborhood of p and σ is a non-vanishing local section of L . Then $[\sigma_0(p) : \dots : \sigma_r(p)] = [f_0(p) : \dots : f_r(p)]$. The map Φ is well-defined (modulo automorphisms of $\mathbb{P}^r$) and its degree equals the degree of $\mathcal{L}$, since there are no base points.

Denote by $V(D) \subset V$ the subspace $\{\sigma \in V \mid \text{div}(\sigma) \geq -D\}$, where $D \in \text{Div}(X)$.

Proposition 2.1. *Let (L, V) be a (base point free) linear system of rank r and degree d on a compact Riemann surface X , with associated map $\Phi : X \rightarrow \mathbb{P}^r$. Then:*

- Φ is injective if and only if $V(-p) \neq V(-q)$ for any distinct $p, q \in X$.
- Φ is an immersion if and only if $V(-2p) \neq V(-p)$ for all $p \in X$.
- Φ is an embedding if and only if $V(-p-q) \subset V$ has codimension 2 for any $p, q \in X$.

Corollary 2.1. *If L is a holomorphic line bundle over X such that $\deg(\mathcal{L}) \geq 2g + 1$, then the map associated to the complete linear system $|L|$ is an embedding.*

Proof. From the proposition, it suffices to show that for all $p, q \in X$,

$$\dim V = h^0(X, \mathcal{L}) = h^0(X, \mathcal{L} \otimes \mathcal{O}_X(-p-q)) + 2 ,$$

since $h^0(X, \mathcal{L} \otimes \mathcal{O}_X(-p-q)) = V(-p-q)$. By Riemann-Roch, we have

$$h^0(X, \mathcal{L} \otimes \mathcal{O}_X(-p-q)) - h^0(X, \Omega_X^1 \otimes \mathcal{L}^* \otimes \mathcal{O}_X(p+q)) = \deg(\mathcal{L}) - 2 + 1 - g$$

and

$$h^0(X, \mathcal{L}) - h^0(X, \Omega_X^1 \otimes \mathcal{L}^*) = \deg(\mathcal{L}) + 1 - g .$$

But by assumption, $\deg(\Omega_X^1 \otimes \mathcal{L}^* \otimes \mathcal{O}_X(p+q)) \leq 2g - 2 - (2g + 1) + 2 = -1 < 0$ and $\deg(\Omega_X^1 \otimes \mathcal{L}^*) \leq 2g - 2 - (2g + 1) < 0$. Thus, the second terms on the left hand side of both equations above vanish. Putting everything together, we obtain the desired relation. $\square$

Corollary 2.2. *If the genus $g \geq 2$, then the map associated to the complete linear system $|3K_X|$ is (the tri-canonical) embedding.*

Proof. It is immediate, since $\deg(3K_X) \geq 2g + 1$ when $g \geq 2$. $\square$

The equivalence between elliptic curves and genus 1 Riemann surfaces can now be revisited in terms of the concepts above:

Proposition 2.2. *Let X be a genus 1 Riemann surface. If L is a holomorphic line bundle of degree 3 over X , then $|L|$ is a base point free linear system of degree 3 and rank 2. Furthermore, the induced map $\Phi : X \rightarrow \mathbb{P}(H^0(X, \mathcal{L})^*)$ is an embedding of X into $\mathbb{P}^2$. Also, once an isomorphism $\mathbb{P}(H^0(X, \mathcal{L})^*) \cong \mathbb{P}^2$ is fixed, there is a homogeneous polynomial $P \in \mathbb{C}[x, y, z]$ of degree 3 such that $\text{Im}(\Phi) \subset \{P = 0\}$.*

Proof. From Corollary 2.1, since $3 = \deg(\mathcal{L}) = 2g + 1$, Φ is an embedding. By the Riemann-Roch formula,

$$h^0(X, \mathcal{L}) - h^0(X, \Omega_X^1 \otimes \mathcal{L}^*) = 3 + 1 - g = 3 ,$$

but since $\deg(\Omega_X^1 \otimes \mathcal{L}^*) = -3$, we have $h^0(X, \mathcal{L}) = 3$, and therefore the complete linear system $|L|$ has rank 2.

It remains to show the existence of the polynomial $P \in \mathbb{C}[x, y, z]$:

We have (L, V) , where $V = H^0(X, \mathcal{L})$ is a 3 dimensional vector space and $\deg(\mathcal{L}) = 3$. Consider the (linear) map

$$\Psi : \text{Sym}^3 V \rightarrow H^0(X, \mathcal{L}^{\otimes 3}) ,$$

that sends degree 3 monomials on variables x, y, z (a basis of V) to their product, which is a section of $\mathcal{L}^{\otimes 3}$.

Observe that $\dim \text{Sym}^3 V = 10$ and by Riemann-Roch, $h^0(X, \mathcal{L}^{\otimes 3}) = \deg(\mathcal{L}^{\otimes 3}) + 1 - g = 9$. Thus, Ψ has a non-trivial kernel, i.e. there exists $0 \neq P \in \text{Sym}^3 V$ such that $\Psi(P) = 0$. Considering x, y, z as the homogeneous coordinates associated to the map $\Phi : p \mapsto [x(p) : y(p) : z(p)]$, it becomes clear that the image of Φ is contained in the zero set of P in $\mathbb{P}^2$. $\square$

We finish this section with a lemma:

Lemma 2.1. *Let X be a compact Riemann surface. If a holomorphic map $f : X \rightarrow \mathbb{P}^1$ has degree 2, then it has exactly $2g + 2$ disjoint critical values.*

Proof. Simply apply the Riemann-Hurwitz formula:

$$2g(X) - 2 = \deg(f)(2g(\mathbb{P}^1) - 2) + \sum_{x \in X} (\deg_x(f) - 1) = -4 + C ,$$

where C is the cardinality of critical points of f , since its local degree on such points equals 2. Thus, f has precisely $C = 2g + 2$ distinct critical points (and values). $\square$

3 Hyperelliptic curves

Definition 3.1. *A compact Riemann surface X of genus $g \geq 2$ is hyperelliptic if and only if there exists a holomorphic map $f : X \rightarrow \mathbb{P}^1$ of degree 2.*

Remark 3.1. *If $g = 2$, then X is always hyperelliptic: $\dim H^0(X, \Omega_X^1) = 2 = \deg(\Omega_X^1)$, so $|K_X|$ is a linear system of the type g_2^1 .*

Theorem 3.1. *The set of isomorphism classes of hyperelliptic surfaces of genus $g \geq 2$ is in bijection with the set of $2g + 2$ distinct points of $\mathbb{P}^1$ modulo automorphisms.*

3.1 Holomorphic Differentials

We know that $h^0(X, \Omega_X^1) = g$ for all genus g Riemann surfaces. But can we exhibit the generators explicitly? It turns out that there is a nice basis for the holomorphic differentials of hyperelliptic surfaces.

From the definition of hyperellipticity and lemma 2.1, the map $f : X \rightarrow \mathbb{P}^1$ has $2g + 2$ critical points. Let $s \in H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2g + 2))$ be a section that vanishes on these points. Notice that the notation $\mathcal{O}_{\mathbb{P}^1}(2g + 2)$ makes sense: The Picard group of the Riemann sphere $\mathbb{P}^1$, $\text{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$ is generated by $\mathcal{O}_{\mathbb{P}^1}(p)$ for any $p \in \mathbb{P}^1$. Locally, s is given by a polynomial of one complex variable and degree $2g + 2$. Taking the pre-image of the graph of s by the map $\phi : \mathcal{O}_{\mathbb{P}^1}(g + 1) \rightarrow \mathcal{O}_{\mathbb{P}^1}(2g + 2)$ (which is just $z \mapsto z^2$ fiber-wise), we have $X \cong Y = \phi^{-1}(\text{graph}(s)) \subset \mathcal{O}_{\mathbb{P}^1}(g + 1)$, where Y is locally expressed by $y^2 = F(x)$, with F a polynomial of degree $2g + 2$.

Proposition 3.1. *Let X be a hyperelliptic Riemann surface, and $f : X \rightarrow \mathbb{P}^1$ a holomorphic map of degree 2 with distinct critical values $p_1, \dots, p_{2g+2} \in \mathbb{P}^1$. Suppose $p_i \neq \infty$ for all i . Consider the embedding of X in $\mathcal{O}_{\mathbb{P}^1}(g + 1)$. The restrictions of the meromorphic 1-forms $\{x^i \frac{dx}{y} | i = 0, \dots, g - 1\}$ to the open set of X given by*

$$\{(x, y) \in \mathbb{C}^2 | y^2 = \prod_{j=0}^{2g+2} (x - p_j)\}$$

coincide with the restrictions of a basis of holomorphic differentials on X .

Proof. Since locally X is given by $y^2 = F(x)$, we may consider, in these coordinates, the (clearly linearly independent) 1-forms $x^i \frac{dx}{y}$, with $i = 0, \dots, g - 1$. By applying exterior differentiation to $y^2 = F(x)$, one gets

$$2 \frac{dy}{F'(x)} = \frac{dx}{y} .$$

on X , where $F'(0) \neq 0$. Thus, the 1-forms $x^i \frac{dx}{y}$ are holomorphic at $(x = 1/s, y)$. At $\infty = (s, t = s^{g+1}y) \in \mathbb{P}^1$, they are still holomorphic: Indeed,

$$\begin{aligned} x^i \frac{dx}{y} &= \frac{(1/s)^i (-ds/s^2)}{t/s^{g+1}} \\ &= \frac{-s^{g+1}}{s^i} \frac{ds}{t} , \end{aligned}$$

which is holomorphic, since the exponent i does not exceed $g + 1$. □

Corollary 3.1. *If X is hyperelliptic, the map associated to $|K_X|$ is factored by the $2 : 1$ map $f : X \rightarrow \mathbb{P}^1$, as in the diagram below:*

$$\begin{array}{ccc} X & \xrightarrow{|K_X|} & \mathbb{P}^{g-1} \\ & \searrow f & \uparrow \eta \\ & & \mathbb{P}^1 \end{array}$$

where $\eta : \mathbb{P}^1 \rightarrow \mathbb{P}^{g-1}$, ($g \geq 2$) is the embedding given by

$$[Z : W] \mapsto [Z^{g-1} : Z^{g-2}W : \dots : ZW^{g-2} : W^{g-1}] .$$

Proof. Fix a basis $\{\frac{dx}{y}, x\frac{dx}{y}, \dots, x^{g-1}\frac{dx}{y}\}$ of $H^0(X, \Omega_X^1)$. Then the map associated to $|K_X|$ has the form

$$\begin{aligned} O_{\mathbb{P}^1}(g+1) \supset X &\rightarrow \mathbb{P}(H^0(X, \Omega_X^1)^*) \cong \mathbb{P}^{g-1} \\ (x, y) &\mapsto [1 : x : x^2 : \dots : x^{g-1}] , \end{aligned}$$

while $f : (x, y) \mapsto x \in \mathbb{P}^1$. □

Corollary 3.2. *If X is hyperelliptic, there exists a unique involution of X with $2g+2$ fixed points.*

Corollary 3.3. *If X is a compact Riemann surface of genus $g \geq 2$ and $f, h : X \rightarrow \mathbb{P}^1$ are holomorphic of degree 2, then there exists an automorphism ψ of $\mathbb{P}^1$ such that $f = \psi \circ h$.*

3.2 The Main Theorem

We finish this survey with the following criterion for hyperelliptic surfaces.

Theorem 3.2. *Let X be a compact Riemann surface of genus $g \geq 2$. The complete linear system $|K_X|$ defines an embedding $X \rightarrow \mathbb{P}^{g-1}$ if and only if X is not hyperelliptic.*

Furthermore, if X is hyperelliptic, then the image of the map defined by $|K_X|$ is isomorphic to $\mathbb{P}^1$.

Proof. Recall that a linear system (L, V) defines an embedding if and only if $V(-p-q) \subset V$ has codimension 2 for any $p, q \in X$.

Suppose that $|K_X|$ does not define an embedding. Then there are points $p, q \in X$ for which

$$h^0(X, \Omega_X^1(-p-q)) = h^0(X, \Omega_X^1) - \delta ,$$

for $\delta \in \{0, 1\}$. Riemann-Roch yields

$$\begin{aligned} h^0(X, \Omega_X^1(-p-q)) - h^0(X, \mathcal{O}_X(p+q)) &= \deg(\Omega_X^1) - 2 + 1 - g \\ &= 2g - 2 - 2 + 1 - g \\ &= g - 3 \end{aligned}$$

Putting things together,

$$\begin{aligned} h^0(X, \mathcal{O}_X(p+q)) &= h^0(X, \Omega_X^1(-p-q)) + 3 - g \\ &= h^0(X, \Omega_X^1) - \delta + 3 - g \\ &= 3 - \delta \in \{2, 3\} . \end{aligned}$$

Thus, $h^0(X, \mathcal{O}_X(p+q)) \geq 2$ and therefore the complete linear system $|p+q|$ yields a $\Phi : X \rightarrow \mathbb{P}^1$ that is $2 : 1$.

Conversely, we already know from Corollary 3.1 that if X is hyperelliptic, then the map $|K_X| = \eta \circ f$ (where f is $2 : 1$ and η is an embedding) cannot be an embedding. □

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